

Name: _____

Question 1

For each variable below, identify its type and if the Poisson or Normal distribution might be appropriate (or neither):

- Adult elephant heights
- Daily number of purchases
- Number of website hits per hour
- Length of human pregnancies
- Salaries of LA residents
- Number of spots on the peel of a banana in a 1 by 1 inch square
- Number of contagious disease cases reported at CSUCI day by day

Question 2

Fill in the table below for the Normal and Poisson distributions.

	Normal distribution	Poisson distribution
Parameters		
Shape of distribution		
Appropriate variable type		
Use case	Measurements (typically)	Counts
Example variable		

Question 3

Suppose people report seeing an average of 0.5 rattlesnakes while hiking a 3 mile loop at Wildwood Regional Park. What distribution could be appropriate to model the number of snake spotted while hiking the three mile trail?

Determine the probability of seeing no rattlesnakes on a 3-mile loop hike at Wildwood Regional Park.

Determine the probability of seeing more than 2 rattlesnakes on a 3-mile loop hike at Wildwood Regional Park.

Question 5

Suppose that a tire manufacturer believes that the lifetimes of its tires follow a normal distribution with mean 50,000 miles and standard deviation 5,000 miles. What is the probability of a tire lasting less than 40,000 miles?